Project 04

Herb Code

Interactive device / User Experience

BRIEF

This project is inspired by the herbal fragrances found in Lady Xin Zhui's Han Dynasty tomb at Mawangdui, Hunan Museum. It aims to make ancient Chinese aromatherapy **accessible and engaging**, helping visitors **better absorb the knowledge it conveys.**

TEAM PROJECT (3 People)

I am mainly responsible for **first-hand research** and **programming** and also participate the second-hand research, ideation and visual design.

DURATION

May. 2025 – Aug. 2025

INSPIRATION

The Hunan Museum showcasing the rich history and heritage of Hunan Province, especially the Mawangdui Han Tombs.



Hunan Provincial Museum

BACKGROUND

Museum Fatigue Phenomenon

"During the process of **visiting exhibitions**, people receive a large amount of information for a long time, gradually experiencing **sensory dullness, decreased attention, cognitive overload and physical fatigue**... ..."

Information overload

There are **hundreds of information display boards**, and each contains more than **200 words of dense text.**

Visitors only read the content of the **first two paragraphs.**

Physical overwalk

Tourists began to seek for seats **one hour later.**

Detour along the edge and abandon the central exhibition area.

The time spent on a single exhibit is only 8 to 15 seconds.

Experiential overlooked

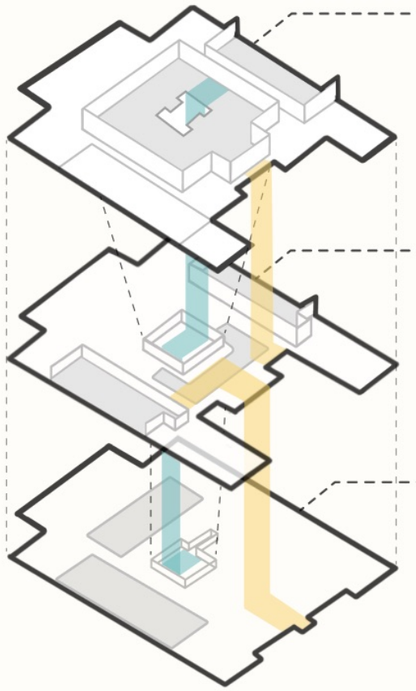
Activities are mainly based on explanations **73%**

Over 31% of the users are family-friendly, but fewer are suitable for children.

RESEARCH

The Hunan Provincial Museum

The museum is mainly organized into three floors, each with a fixed theme and divided into several smaller exhibition zones.



Mawangdui Han Dynasty Tombs Exhibition

3rd Floor

The **fully preserved tomb structure** as well as a **wealth of funerary items** serve to fully embody **everyday life, Craftsmanship and funeral concepts** during the Han Dynasty.



Hunanese--Exhibition of Hunan Culture

2nd Floor

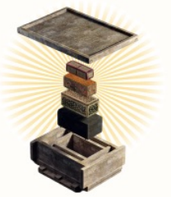
Told from the perspective of a Hunan native, this narrative **explores the history and culture of Hunan**, and reflects on **how the land and its history have shaped the way we live.**



Coffin of the Tomb & Hall

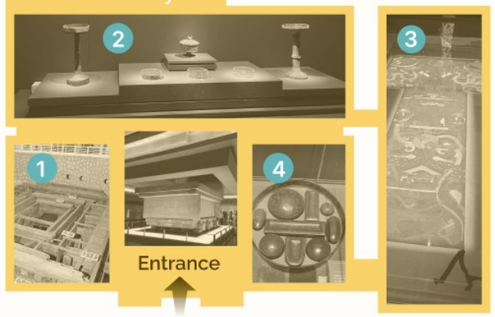
1st Floor

Showcases **the body of Lady Dai (Xin Zhui), her coffin structure and preservation techniques**, which reflect the advanced funerary craftsmanship of two thousand years ago.



Mawangdui Han Dynasty Tombs Exhibition

Exhibition Layout



1 ---------- Epic Discovery
Shows the detail of the tomb's history & structure.

2 ---------- Life & Art
Daily life items and burial goods of the tomb owner.

3 ---------- Manuscript Treasures
Bamboo slips and silk manuscripts documenting valuable cultural knowledge.

4 ---------- Dream of Immortality
Secrets of thousand-year-old undecayed tombs.

Visitor Favorites

Visitors spend over **65% of tour time** in this area.

The foot traffic accounts for about **60% of total visitors.**

Museum Information Display Formats

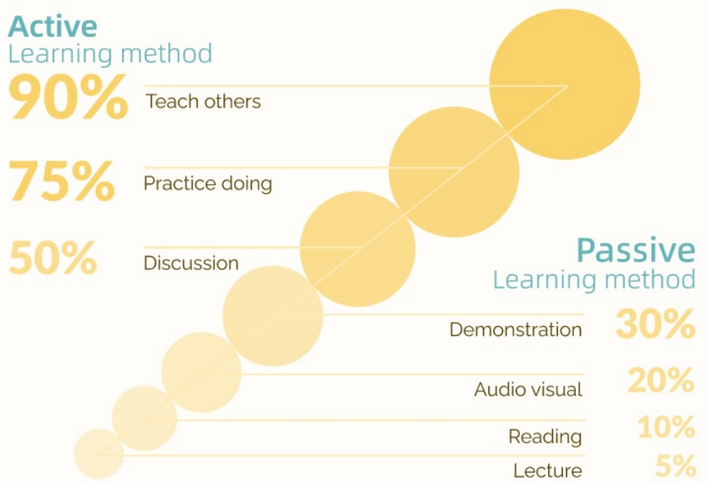
Four different information delivery methods in the museum and **their proportions**

TEXT -- 60% | IMAGE -- 30% | VIDEO -- 7% | WEB -- 3%

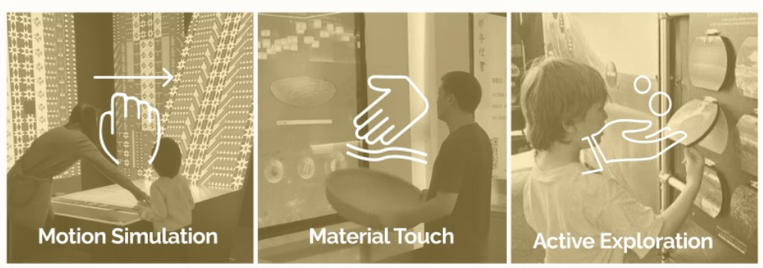
Visitors spend an average of **19 min** at interactive exhibits, **13 min** at static ones.

Learning Pyramid

The **average learning retention rate** of learners over two weeks when adopting **different learning methods**;



Presentation Methods Elsewhere



Takeaways

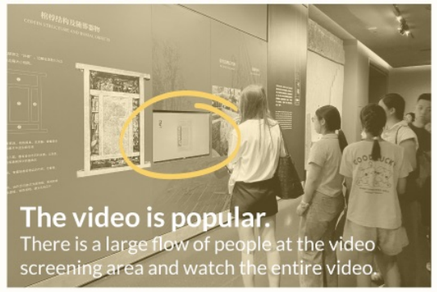
Museums still rely on traditional ways to display information, which can overload visitors and cause physical and mental fatigue.

OBSERVATION

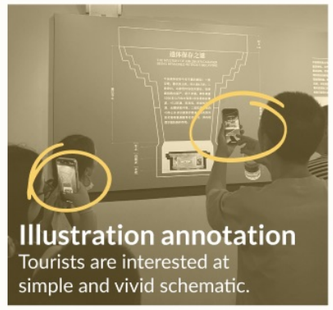
Visitor Behavior During the Tour

Observe the behavior of visitors in the **cultural relic display cabinets, information display area and interactive installation area** of the exhibition zone

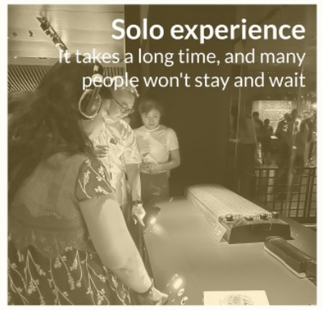


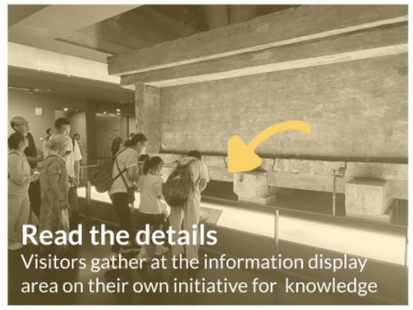












INTERVIEW

Methods

Semi-structured interviews

16 respondents

Methods

Age 35 — Ms.Chen

"The display of information **could be more visual**, and add some **interactive games**. The text and images can be presented as **a dynamic storyline**."

age 55 — Mr.Deng

"The **main functions** of the components could be **explained more clearly**. An **animation** showing how the incense was burned and used would help."

Age 23 — Ms. Huang

"When I want to dive deeper, the **information is insufficient**, and **the explanations are rather superficial**, making it **hard to fully engage**."

Age 19 — Mr.Liu

"The images and text explanations of the exhibits are clear, but I hope there could be more **stylized presentations**; otherwise, it can **feel quite dull**."

Age 31 — Ms.Liang

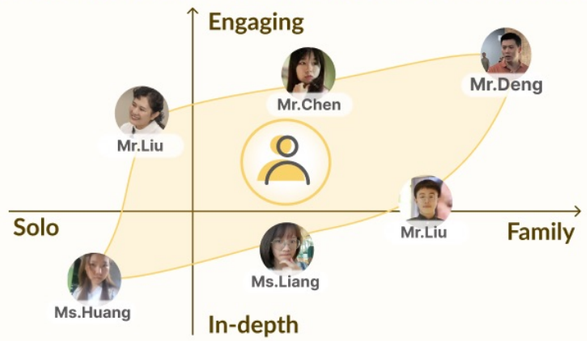
"The ancient artifacts have **become deformed and don't match their original shape**. Details **aren't clear from text alone**, so more visuals are needed."

Age 43 — Ms.Lin

"I think it would be great to **set up a scenario-based Q&A session after the tour**, which can help visitors remember what they've seen."

User Matrix

I categorized six interviewees into a user matrix to identify the most common typical visitors in the museum.



Persona



About

A culturally engaged family, often visits historical sites and museums together, enjoying lighthearted trips.

Core need

- Family-shared tour experience.
- Information accessible to all ages.

Pain point

- Prone to fatigue in the later part of visit.
- Family members have different interests.

ANALYSIS

Findings

- Prefer concise visuals
- Skip complex information
- Frequent chatting
- Prefer multi types input
- Want deeper knowledge
- Interaction does not suit all
- Don't finish full process
- Adore hands-on activities
- Interactive devices affected by environment

Insights

Present information with a balance of vividness and depth.

Visitors prefer both simple, intuitive visuals and in-depth, complex knowledge. Design should balance clarity with depth, offering multiple formatsto suit different audiences.

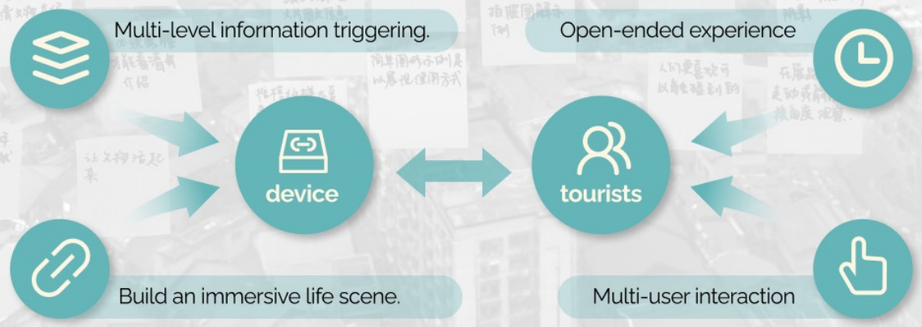
Interaction should balance accessibility and flow.

Current interactive devices are not suitable for all visitors. They should be more inclusive, easy to engage with, and seamlessly integrated into the visiting route.

Enable effective Multiple types and hands-on experiences.

Interactive installations combining visual, tactile elements, as well as hands-on experiences will help make knowledge more memorable and emotionally engaging.

OPPORTUNITY



IDEATION

Idea Selection

Ideas are around how **abstract culture and bodily experience** could be transformed into **a perceptible and participatory system.**

Then I **evaluated** them to select the final direction.

	Assembly of coffin structure	Gauze Gown Virtual Try-On	Herbal Medicine Combination
			
Multi-level information triggering.	✓	✗	✓
Multi-person collaborative interaction	✓	✓	✓
Open-ended experience	✗	✓	✓
Build an immersive life scene	✗	✗	✓

CONCEPT

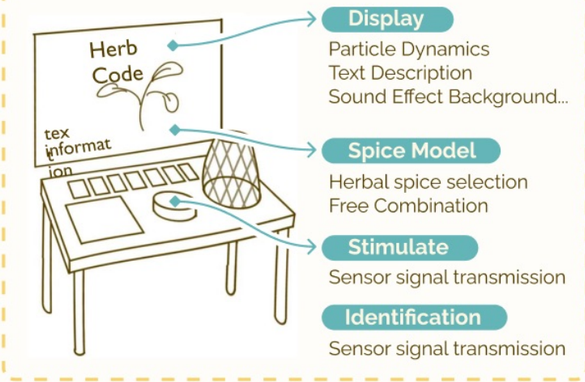
Mawangdui spice

Herbal spices and incense methods preserved from the Han Dynasty

Practice doing have 70% learning retention rate

Using tech-driven interfaces to **visualize traditional content.**

Sketch



EXPLORATION

Types of Herbal Spices



Incense-burning Process



Considerations

- Real dried spices are fragile, and may be contaminated by visitors' touch

  Sealing them in transparent **UV resin blocks** can preserve their appearance and color intact.

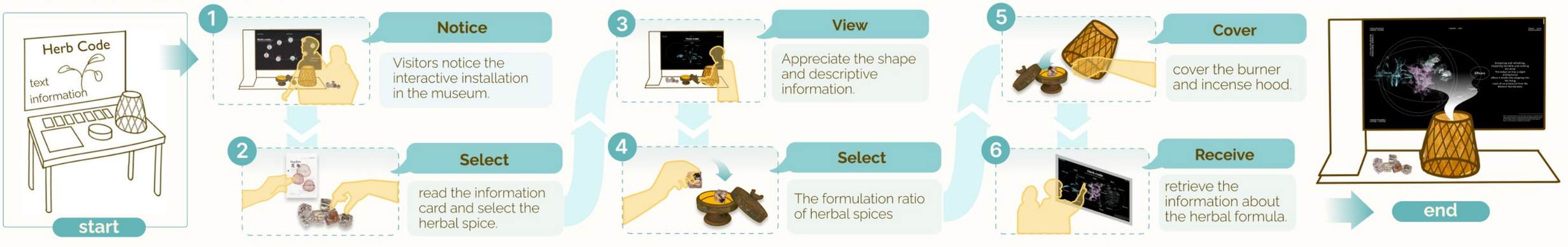
- Visitors cannot clearly distinguish the effects of different formulas

  Using **combinations of spices** to conveys the principles and cultural background of traditional Chinese medicine .

- Traditional exhibits have dense text descriptions and lack visual hierarchy

  Remove lengthy text entirely and replace it with intuitive visual language.

EXPERIENCE FLOW



PROTOTYPE

01 Arduino INPUT

Arduino enables the creation of multi-sensory interactive demos using various sensors.

Components

1. **Arduino board** — Controls and processes input/output.
2. **RFID-RC522 module** — Reads and identifies RFID tags for object detection.
3. **Atomizer module** — Produces fine mist for effects.
4. **RGB LED** — Emits colored light.
5. **Breadboard** — Provides a reusable platform for circuit.

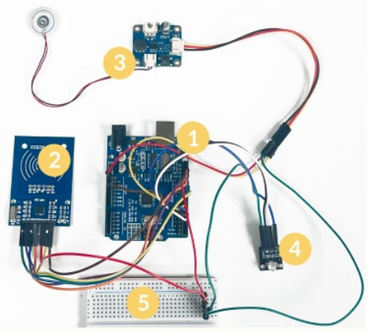
RFID card (NFC) select



A high-frequency RFID chip with a storage capacity of 1024 bytes.

MIFARE Classic 1K

Assembly

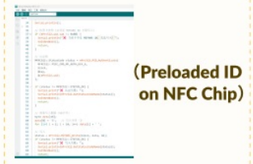


Code

Load libraries
Communicate with and control the MFRC522 RFID module.

Initialize Hardware
Initialize serial, SPI, RFID, and outputs; turn mist off.

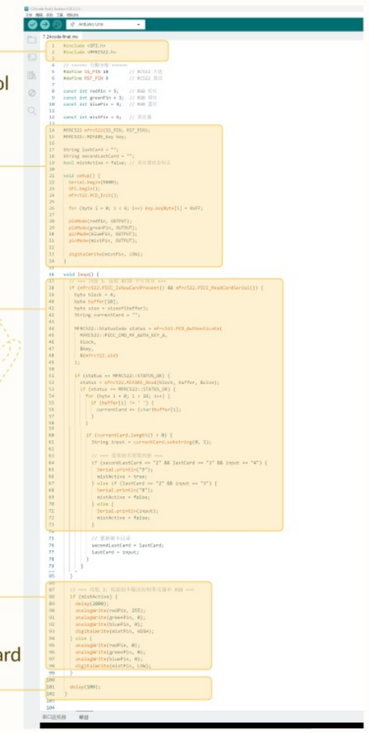
Read RFID Card Data
Read block 4 of RFID card and get character as ID.



(Preloaded ID on NFC Chip)

Signal control
Control the LED & atomizer based on signals from RFID card

Refresh Interval



02 Touchdesigner OUTPUT

TouchDesigner is a node-based visual programming tool for creating real-time interactive and multimedia experiences, widely used in art, performance, and installations.

Concept ---------- Coding ---------- Visual Outcome

Han dynasty

The spices in the museum have been preserved for thousands of years.



All in fragmented form

Nowadays

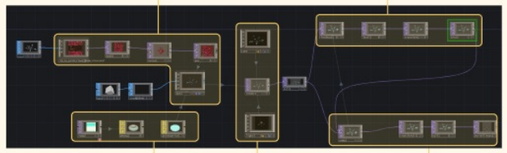
Before processing and preservation, the nine Chinese herbs served as herbal spices.



Shapes varied naturally

Visual effects production

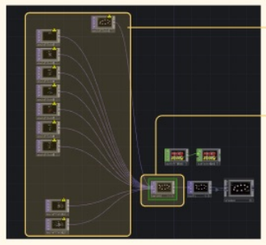
particle & motion effects — **glow Trail effecct**



rendering — Pre-made 3D model

output — final display

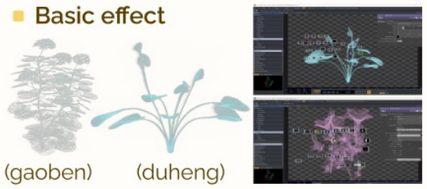
Hardware communication



file importing — Pre-made & numbered files

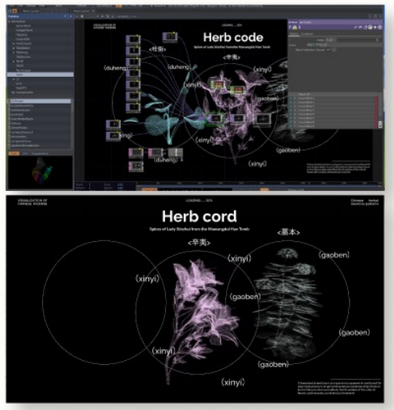
Receive signal switch files — Controlled by **Arduino** then **output**

Basic effect



(gaoben) (duheng)

Final interface



Fabrication



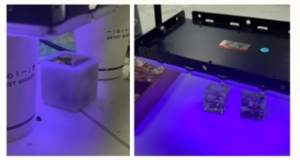
Making spice cubes with UV resin



Bamboo-silk incense cover



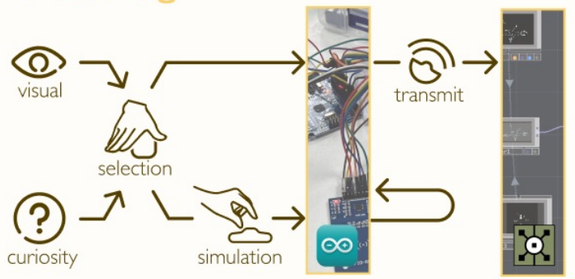
Make spice and herb cards



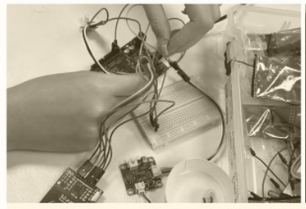


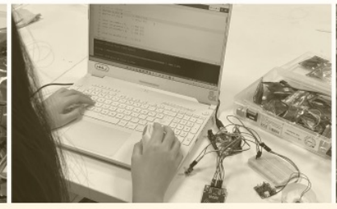


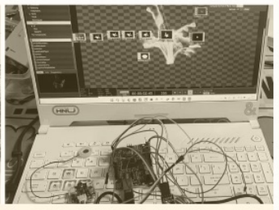
Testing



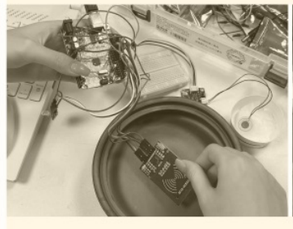








01 Debugging code and hardware communication

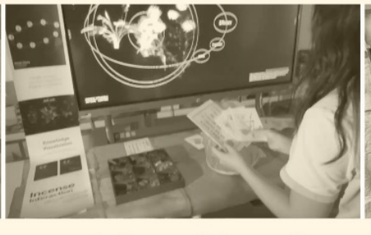






02 Assembly of hardware and props; interactive device construction

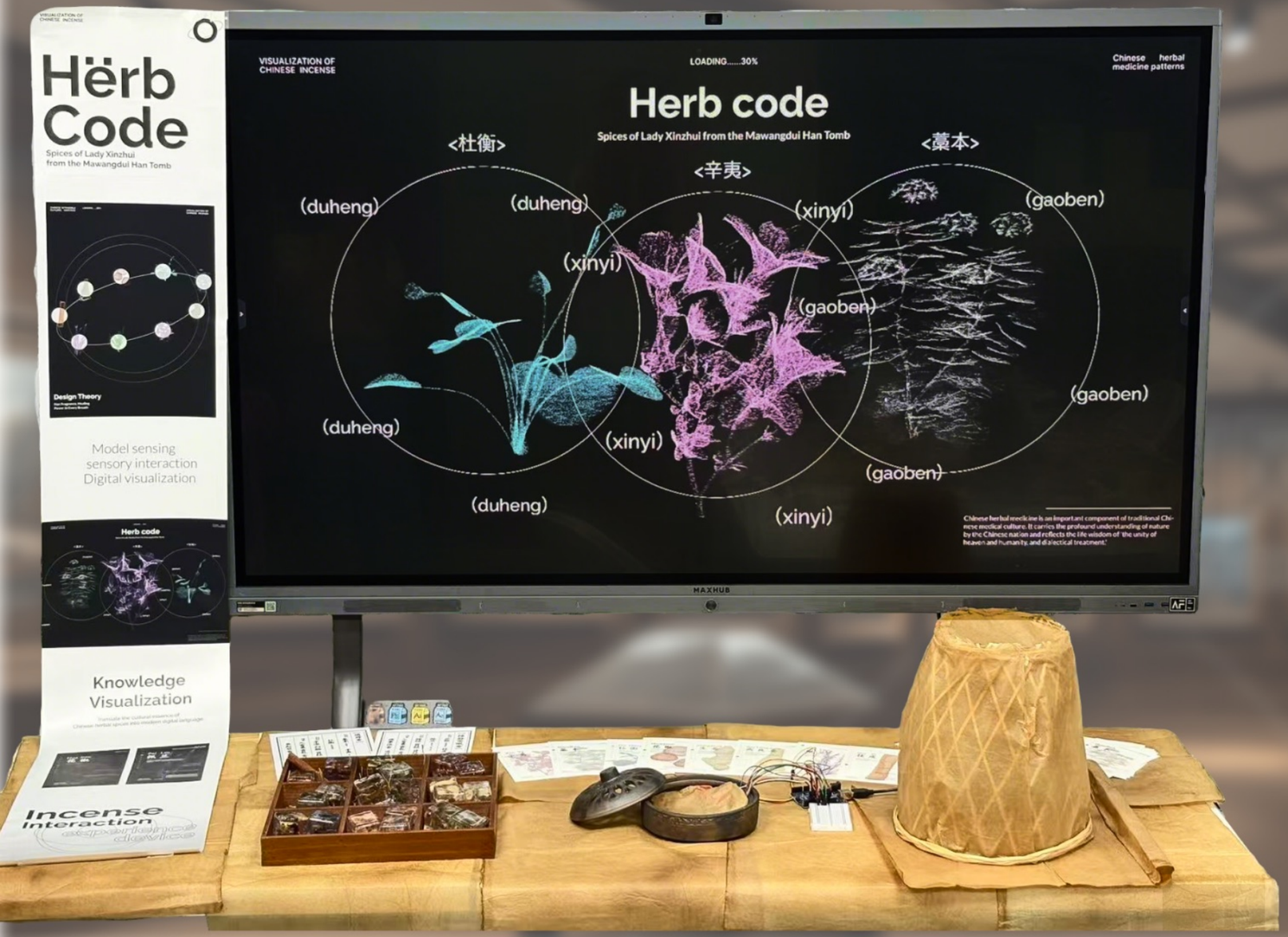






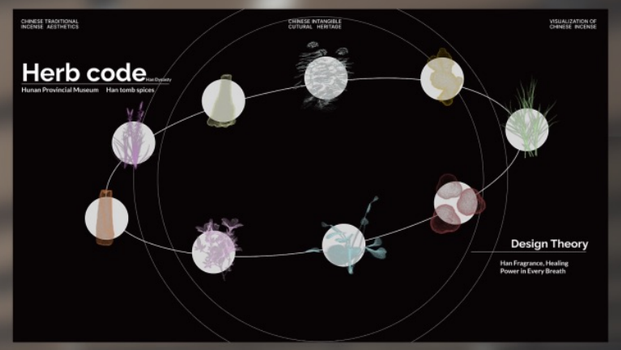
03 Overall device completion and decoration

FINAL OUTCOME

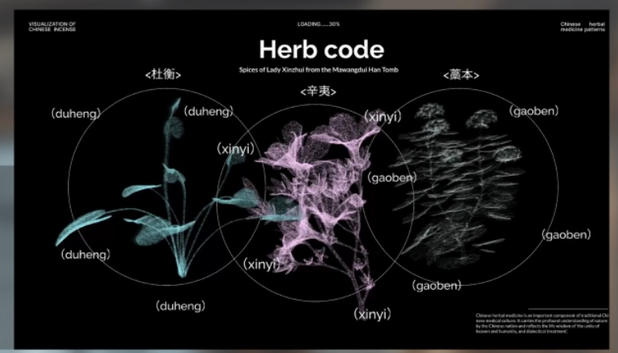


■ Visualization

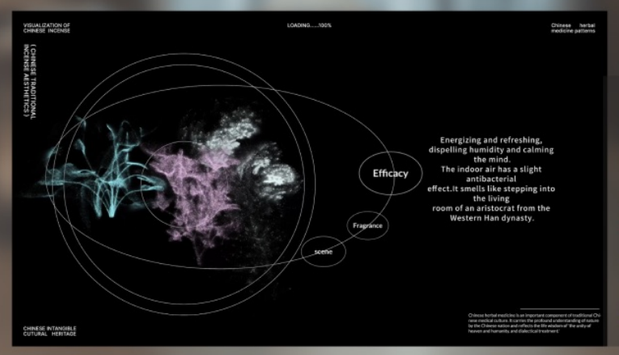
Stylization and visualized presentation of herbal spices within the installation.



Stage 1: Idle State
A dynamic exhibition displaying all types of herbs, allowing users to observe and browse their various forms.



Stage 2: Herb Selection
Users interact with herb blocks, assembling the particles to reveal the actual forms of the herbs.



Stage 3: Prescription Formation
By combining the selected herbs, a prescription overview is generated, including its description and therapeutic effects, deepening the user's understanding and knowledge.

■ DIsplay





REFLECTION

This covered Han incense basics well, but it could go further...

Yeah, I will try to swap Arduino for something **more adaptable**, **embed the components** better, design **cooperative modes**, and use visuals to highlight those proportion differences **more vividly**!